

IMPACT ANALYSIS

CR-009: Student Dashboard Course Display and Enrollment

1. Change Summary

CR-009 enhances the Student Dashboard by adding an **Available Courses** section. Courses available in the system will be retrieved and displayed to authenticated students.

Students will be able to view relevant course information and enroll directly in a selected course from the Student Dashboard. The existing administrator course management functionality, where available, will not be changed by this CR.

The change mainly affects the Student Dashboard and requires integration with the course and enrollment functionality required for the student enrollment workflow.

2. Impact Assessment

Area	Impact	Assessment
Frontend	High	Student Dashboard will be enhanced to display available courses, course details, and an Enroll button.
Backend	Moderate	Backend course and enrollment functionality may need to be integrated or introduced to support the Student Dashboard workflow.
API	Moderate	Course retrieval and enrollment endpoints will be required for communication between the frontend and backend. Existing suitable endpoints will be reused where available.
Database	Moderate	Course and enrollment data must be stored and retrieved correctly. Existing tables will be reused where available; otherwise, the required database structures will be introduced as part of implementation.
Authentication	None	Existing student authentication remains unchanged.
Authorization	Low	Existing student authorization must continue to restrict enrollment operations to authenticated students.
Performance	Low	Courses will be loaded through the API and displayed on the dashboard while avoiding unnecessary repeated requests.

CRUD	Low	Any existing administrator course CRUD functionality must continue to operate correctly without being negatively affected by CR-009.
Deployment	Low	No major deployment changes are expected. Any required database schema setup will be performed as part of implementation.

3. Frontend Impact

The main affected area is the **Student Dashboard**.

Potential frontend changes include:

- Adding an **Available Courses** section.
- Loading available courses from the backend.
- Displaying relevant course information.
- Adding an **Enroll** button for available courses.
- Displaying enrollment success messages.
- Displaying enrollment error messages.
- Handling loading and empty-course states.
- Adding responsive styling for course display.
- Preserving the existing student profile information.

The existing Student Dashboard profile section must continue to function correctly after the change.

4. Backend, API and Database Impact

CR-009 requires the Student Dashboard to communicate with the backend for course retrieval and student enrollment.

The backend must provide the required functionality to:

1. Retrieve available courses for authenticated students.
2. Return the required course information.

3. Accept enrollment requests from authenticated students.
4. Validate the selected course.
5. Store the student's enrollment correctly.
6. Return appropriate success or error responses.

Where suitable course and enrollment functionality already exists, it will be reused. Where the current system does not provide the required functionality, the necessary backend components and database structures will be introduced as part of CR-009.

The expected data areas are:

1. **courses**
2. **enrollments**
3. Existing **student/user-related tables**

The change must ensure that course and enrollment data are correctly connected to the authenticated student.

Any existing course or enrollment functionality must continue to operate correctly without negatively affecting other system functions.

5. Security Impact

The existing authentication and authorization mechanisms must remain unchanged.

The enrollment feature must:

- Allow only authenticated students to enroll.
- Preserve existing student authorization controls.
- Prevent unauthenticated users from performing enrollment requests.
- Use the existing authenticated API mechanism.
- Not bypass existing backend authorization.
- Not expose unnecessary student or course-related sensitive information.
- Preserve existing authentication tokens and security controls.

The change must not weaken the security of the existing Student Management System.

6. Course and Enrollment Functional Impact

CR-009 must ensure that the Student Dashboard correctly supports the course viewing and enrollment workflow.

Particular attention is required for:

- Displaying available courses.
- Showing the correct course information.
- Selecting the correct course for enrollment.
- Sending the correct course identifier during enrollment.
- Validating the selected course.
- Preventing invalid enrollment requests.
- Displaying an appropriate success message after successful enrollment.
- Displaying an appropriate error message when enrollment fails.
- Preventing unauthorized enrollment attempts.

The Student Dashboard enhancement must not modify or interfere with any existing administrator course management functionality.

7. Performance Impact

The expected performance impact is **low to moderate**.

Courses will be retrieved through the backend API and displayed on the Student Dashboard.

The implementation should avoid unnecessary repeated API requests and should load course information efficiently.

The number of displayed courses should be appropriate for the current expected system size.

If the number of courses becomes significantly larger in the future, pagination, server-side filtering, or other optimization techniques may be considered as a separate change.

8. Testing Impact

Manual and functional testing should cover:

- Student Dashboard loading.
- Existing student profile information.
- Display of available courses.
- Correct course title, category, level, duration, and price.
- Course selection.
- Successful student enrollment.
- Enrollment success message.
- Failed enrollment attempt.
- Enrollment error message.
- Correct course ID being used for enrollment.
- Invalid course enrollment handling.
- Unauthenticated access restrictions.
- Student authorization.
- Existing administrator course management, where available.
- Existing authentication functionality.
- Responsive display of available courses.
- Empty-course scenario.
- Regression testing of existing Student Dashboard functionality.

9. Risk Analysis

Risk	Impact	Mitigation
Incorrect course selected for enrollment	High	Preserve and use the correct course ID when sending enrollment requests.
Enrollment request fails	High	Validate API requests and provide appropriate error handling and user feedback.
Unauthorized user performs enrollment	High	Preserve existing authentication and authorization controls and enforce authorization on the backend.
Existing course management functionality is affected	High	Perform regression testing of any existing course CRUD functionality.
Course information is displayed incorrectly	Medium	Verify frontend data mapping against the backend course response.
Duplicate or invalid enrollment	Medium	Apply backend enrollment validation and prevent invalid enrollment requests.
Student Dashboard becomes slow	Medium	Avoid unnecessary API requests and optimize course rendering.
Responsive UI problems	Low	Test the dashboard on different screen sizes and devices.

10. Overall Impact and Recommendation

Overall Impact: Moderate

CR-009 is primarily a **Student Dashboard enhancement** that provides students with access to available course information and allows them to enroll directly from the Student Dashboard.

The main impact is on the frontend Student Dashboard, with backend and API integration required for course retrieval and enrollment processing. Course and enrollment database structures will be reused where available or introduced where required to support the functionality.

The change does not aim to modify administrator course management functionality. Existing authentication and authorization mechanisms must be preserved to ensure that only authenticated and authorized students can enroll.

The change is technically feasible and recommended for implementation, subject to formal approval and successful functional, security, and regression testing.

Key implementation considerations:

- Available courses must be displayed correctly on the Student Dashboard.
- The correct course ID must be used during enrollment.
- Only authenticated and authorized students must be allowed to enroll.
- Existing course management functionality must continue to work correctly.
- Existing Student Dashboard profile functionality must remain unchanged.
- Appropriate success and error messages must be provided.
- Course and enrollment data must be stored and retrieved correctly.
- Security controls must remain enforced throughout the enrollment workflow.